New York University — Tandon School of Engineering

Department of Technology Management and Innovation

IE-GY 9113B — Systems Integration: From ERP to Agentic AI

Study Note — Week 5: Scaling RPA and the Center of Excellence

Module 2: The Automation Ladder | Friday, October 2, 2026

Readings for this week	
Primary case	Robotic Process Automation at Telefónica O2 — Mary Lacity and Leslie Willcocks, *MIS Quarterly Executive*, 2016; the peer-reviewed account of a pioneering programme that scaled RPA through a small in-house team and a Back-Office-owned operating model. Read the open-access working paper (LSE Outsourcing Unit 15/02): https://eprints.lse.ac.uk/64516/1/OUWRPS_15_02_published.pdf
Supporting	Orange Spain, 'Robot Factory' — a UiPath customer case study. This is vendor marketing, not independent research: treat every figure in it as a claim the vendor is making, not a verified fact. We read it only to see the shape of a second telecom's Center-of-Excellence growth arc, and to practise reading a vendor case against an independent one: https://www.uipath.com/resources/automation-case-studies/orange-spain-telecom-rpa
Go deeper (optional)	Deloitte, *Automation with intelligence* (2022). The 7th Global Intelligent Automation survey (479 executives, 35 countries). Read it for the barriers that stop programmes scaling and for what a Center of Excellence is for. Open access: https://www.deloitte.com/us/en/insights/topics/talent/intelligent-automation-2022-survey-results.html Tella, Brinker and Pezzini, *The New Automation Mindset*, Ch. 4–5 (the scale mindset and orchestration); Westerman, Bonnet and McAfee, *Leading Digital*, Ch. 3 (exploiting the power of core operations). Reference, not required reading — the note teaches the ideas directly and points you to the source at the exact place it uses each one.
Before next session	Lab 2 is in Week 6 and it runs on Make (free tier). Create your Make account this week so the build time next week is spent building, not signing up.

You Named the Stall Last Week — This Week You Build the Thing That Prevents It

This course moves through four stages — integrate, automate, make intelligent, make agentic — and everything you build sits on the first. You are still on the second stage, still climbing the automation ladder inside Module 2, and still, in project terms, in the Diagnose phase you have been in since Week 1: you are working out what your company should do about automation before you build the enterprise version of it. But the question has changed shape. Week 4 taught you to

pick a single good candidate — the repetitive, rules-based, stable step that deserves a bot — and it left you holding a warning it promised to come back to. A bot works the screens, not the plumbing, so it breaks the moment a screen it depends on moves; and Week 4 said, in as many words, remember that failure mode, because it returns in Week 5 as the reason RPA programmes stall at scale. This is Week 5. Here is where that failure mode grows from a footnote about one bot into the central problem of running hundreds.

Week 4 also handed you a second thread. When it zoomed out to the market, Deloitte's 2022 survey reported that the hardest barriers to scaling automation were process fragmentation, a lack of clear vision, a lack of IT readiness, and resistance to change (Deloitte, 2022) — and you were asked to notice that not one of them is the technology. Week 4 named the organisational answer to those barriers and then deliberately withheld it: the Center of Excellence, the thing that governs the jump from a pilot to an enterprise programme. It even pointed the last of Week 4's study questions straight at it — what is the single biggest governance risk of deploying RPA without a Center of Excellence? This week delivers the answer you were told to expect. By the end of it you should be able to explain why programmes die between ten bots and a thousand, design the operating model and governance that keep them alive, and say precisely where a human still belongs even in a system built to run without one.

Keep the three threads in your ear, because this module leads with one of them and this week is no exception. The Customer thread is taking an order and serving the person as one coherent promise; the Operations and Supply Chain thread is planning, inventory, and synchronization, so a company can deliver what it promised; and the Cash Management thread is how fast and how reliably the backbone turns activity into cash — the thread that matters because cash flow is the most common way businesses fail. Scaling RPA lives first on Operations and Supply Chain, because that is where the repetitive, rules-based work is generated and where a large programme finds most of its bots. But watch it land on the other two exactly as Week 4 did: when Telefónica O2 drove a routine service request from days down to minutes, that compressed cash conversion (Cash) and removed the customer's need to phone and chase (Customer) in the same motion. One programme, three threads.

And keep the four archetypes in view, because a Center of Excellence looks different at each scale: a consumer-packaged-goods manufacturer running sales-and-operations planning on SAP; a mid-market manufacturer running order-to-cash on NetSuite; a Series-B startup running order-to-cash across Stripe, Shopify, and QuickBooks; and a fifty-person services firm living in HubSpot and Notion. The primary case this week, Telefónica O2, is a very large enterprise, and its programme has a shape a Fortune-500 board would recognise. Part of your work is to translate that shape down to your own company's scale — because the governing idea holds at every size, even when the fifty-person firm's 'Center of Excellence' is one person and a written rule.

This week answers the question Week 4 left open. Week 4 chose one good bot and flagged two things for later: the brittleness that breaks a bot when a screen moves, and the Center of Excellence that Deloitte's non-technical barriers point to. Week 5 develops both. You are still on the automate rung and still in the Diagnose phase — now designing how

automation is run across a company, not building a single bot. Operations and Supply Chain leads; Cash and Customer follow.

The Week's Question: Your Pilot Worked — So Why Do Most Programmes Die on the Way to Scale, and Who Keeps Them Alive?

Here is the question the whole session turns on. The RPA pilot is one of the most reliably successful demonstrations in enterprise technology: pick a high-volume, rules-based step, point a bot at it, and within weeks you have a working automation and a happy sponsor. Week 4 taught you to run exactly that play. The trouble is what happens next. The pilot that dazzled at three bots behaves nothing like a programme at three hundred. The same surface-level fragility that made one bot cheap to build makes a thousand bots expensive to keep running; the same easy path that let one team stand up a bot in a fortnight lets forty teams stand up four hundred bots that nobody owns; and the infrastructure that carried the pilot quietly buckles when the load multiplies. So the question is two-sided, just as Week 4's was. Why does a programme that works brilliantly small break as it grows — and what organisation, deliberately built, keeps it working as it scales? Answering that is the entire skill of the week, and it is exactly what your Board Memo §4 will ask you to design for your own company: not another list of bots, but the operating model that would let a company run them.

Ten Bots Is a Demo; a Thousand Bots Is an Operation

Start with why programmes stall, because the failure is not random and it is not about the tool being bad. It follows directly from what a bot is. Recall the definition from Week 4: a software robot operates a company's existing applications through their user interface — driving the same screens a person would, reading fields, clicking buttons, copying a value out of one screen and typing it into another by a fixed script. That surface-level approach is the source of both RPA's appeal and its fragility. At the scale of one bot, the fragility is a manageable maintenance chore. At the scale of hundreds, it compounds into the thing that kills programmes. Four failure modes do most of the killing, and you should be able to name them the way you name the three candidate tests.

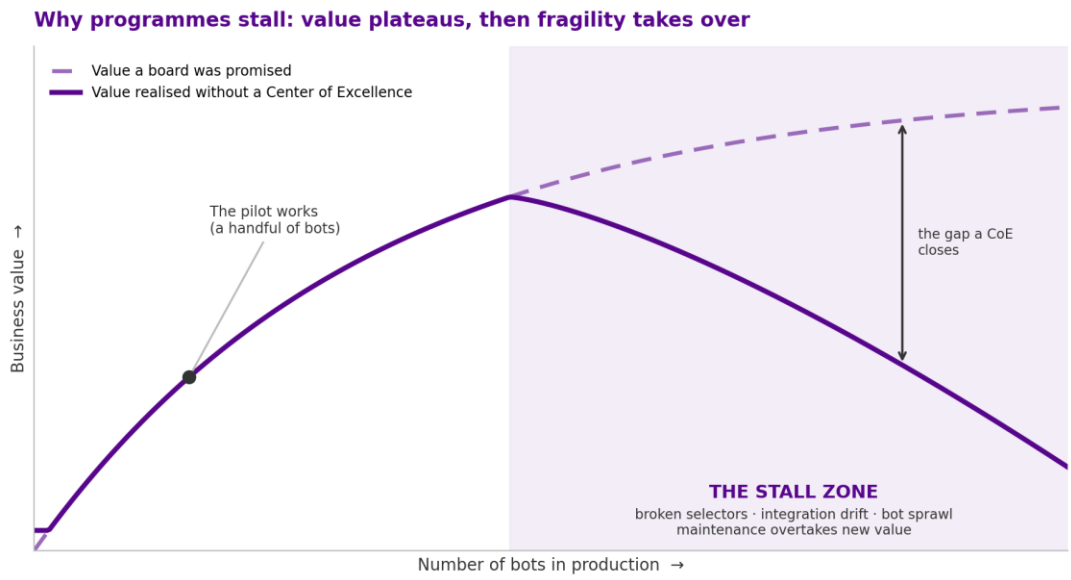


Figure 1. The value a pilot promises keeps rising with bot count; the value a programme without a Center of Excellence actually realises plateaus and then falls, as maintenance and fragility overtake new work.

The first and most concrete is selector instability, and it is worth slowing down on because it is one of this week's study questions and because it makes the abstract word 'brittle' into something mechanical you can picture. A bot does not see a screen the way you do. To click the right button or read the right field, it needs an address for that element — a selector, a description of where the element sits in the structure of the page or window: this button, in this panel, with this label, at this position. When the bot is built, those selectors are recorded against the screen exactly as it looks that day. Now the vendor of that application ships a routine update. A field is renamed, a button moves into a new menu, a pop-up is added to confirm an action, a column is inserted into a table. To a human, none of this is disruptive — you glance at the new layout and carry on. To the bot, the selector it was given no longer points at anything, or points at the wrong thing, and the script fails. That is selector instability: the bot's dependence on the screen looking exactly as it did when it was built, and its inability to improvise when the screen changes underneath it. An 'unexpected system update' is not exotic; it is a Tuesday. The customer never approved it, nobody told the automation team, and one morning a bot that ran flawlessly for a year is silently doing nothing — or worse.

'Or worse' is the half that matters most, and Telefónica O2's own programme supplies the archetype. A bot follows its rule with perfect confidence even when the rule has become wrong for the case in front of it — it does not pause, does not doubt, does not notice the thing a person would have caught out of the corner of their eye. When Apple opened pre-orders for a new iPhone, some of O2's customers, in their enthusiasm, ordered the same phone several times. A human clerk would have recognised that a single customer really wanted a single phone. The bot did not: it followed its instructions and shipped the duplicates (Lacity and Willcocks, 2016). The lesson O2 drew is exactly the scaling lesson — that automating a step which 'obviously' had sufficient rules still forced them to add explicit common-sense checks that a person had supplied for free. Multiply that across hundreds of bots and thousands of edge cases and you see the shape of the

maintenance burden: every bot is a promise that the world will keep looking the way it did the day the bot was born, and the world does not keep that promise.

The second failure mode is integration drift, which is selector instability spread across time and across a whole estate. A single bot bridges two systems at the surface. A programme bridges dozens of systems with hundreds of bots, and every one of those systems is independently evolving — patched, upgraded, reconfigured, occasionally replaced — on its own schedule, by teams who have never heard of your bots. The surface each bot rides on is slowly drifting out from under it. What was a stable seam becomes a moving one, and the cost of keeping the estate running rises not with the number of bots but with the number of connections between bots and changing systems, which grows far faster. This is the mechanism behind the plateau in Figure 1: past a certain size, a programme without a governing function spends more and more of its capacity re-fixing bots that broke and less and less building bots that add value, until maintenance overtakes new work and the realised value bends downward even as the bot count keeps climbing.

The third failure mode is infrastructure brittleness, and O2 lived it in a way that is almost too neat. The company chose to run its bots on virtual machines, with a ‘lead’ machine orchestrating the others. That arrangement worked cleanly at around twenty robots. When the robot count quadrupled, it collapsed; the head of the programme described the mismatch memorably — *“It was like driving a Ferrari with a lawn mower engine.”* The point is not the specific technology; it is that the platform which comfortably carries a pilot is not the platform that carries a programme, and the two look identical right up until the load crosses a threshold and the smaller one fails. O2 needed roughly sixteen weeks to re-engineer the infrastructure — moving servers and databases closer together to cut latency — before the scaled programme ran properly (Lacity and Willcocks, 2016). A pilot proves the automation logic; it does not prove the plant that will run it at ten times the volume.

The fourth failure mode is the one that has no technical cause at all, and it is the bridge to the rest of the week: bot sprawl, the uncontrolled proliferation of automations that nobody owns. When building a bot is easy and there is no sanctioned path for doing it well, teams build their own — quietly, off to the side, to solve a problem they could not get onto anyone’s roadmap. The automation literature has a name for this, and Tella, Brinker and Pezzini describe it plainly in the reading: shadow IT, which in its worse forms becomes rogue IT, where business data is moved and transformed in unsanctioned ways that no one is watching (Tella, Brinker and Pezzini, 2023). Each shadow bot is individually reasonable and collectively a liability — undocumented, unmonitored, duplicated across teams, and quietly accreting the technical debt that lands on IT’s backlog when one of them fails. This is precisely why Deloitte’s most-cited barrier to scaling is process fragmentation (Deloitte, 2022): a scatter of local, unconnected automations is not a programme, it is fragmentation with bots on top. Note where this points. The first three failure modes ask for engineering discipline. The fourth asks for an organisation — someone whose job is to make sure automation is built well, owned, and watched. That someone is the subject of the rest of this note.

Why programmes stall	What goes wrong at scale	Anchored exemplar
Selector instability	A routine system update moves or renames a screen element; the bot's address for it no longer resolves and the script fails — silently or loudly	O2: a bot shipped duplicate iPhones because it followed its rule without the common sense a person supplies
Integration drift	Dozens of systems evolve on their own schedules; the surfaces hundreds of bots ride on drift, and maintenance grows faster than the bot count	O2's estate of 15 core processes across many systems — each change a chance to break a brittle seam
Infrastructure brittleness	The platform that carries a pilot buckles when load multiplies; the failure is invisible until a threshold is crossed	O2's lead virtual machine — fine at ~20 robots, imploded when the count quadrupled; ~16 weeks to re-engineer
Bot sprawl (shadow automation)	With no sanctioned path, teams build their own unowned, unmonitored bots — fragmentation with bots on top	Shadow IT hardening into rogue IT (Tella et al., 2023); Deloitte's No. 1 barrier is process fragmentation (2022)

A pilot proves the logic; it does not prove the programme. Bots stall at scale for four reasons: selector instability (a screen changes and the script breaks), integration drift (many systems evolving under many bots), infrastructure brittleness (the pilot's platform buckles under load), and bot sprawl (unowned shadow automations). The first three demand engineering discipline; the fourth demands an organisation. That organisation is the Center of Excellence.

A Company Has No Brain for Automation, So It Builds One

A single successful bot creates a demand it cannot satisfy. Once one team has proved that automation works, every other team wants theirs, and the company faces a choice it usually does not recognise as a choice. It can let each team fend for itself — the road to bot sprawl — or it can build a standing function whose job is to make automation a governed capability rather than a scatter of one-off scripts. That function is the Center of Excellence, or CoE: a small, permanent team that owns how automation is done across the company. It is not a bigger bot factory. It is the organisational answer to the four failure modes of the previous section, and it is the thing Week 4 promised and Deloitte's non-technical barriers point straight at.

Be precise about what a CoE actually does, because 'Center of Excellence' is the kind of phrase that can mean everything and therefore nothing. Five functions do the real work, and you should be able to design each one for your own company.

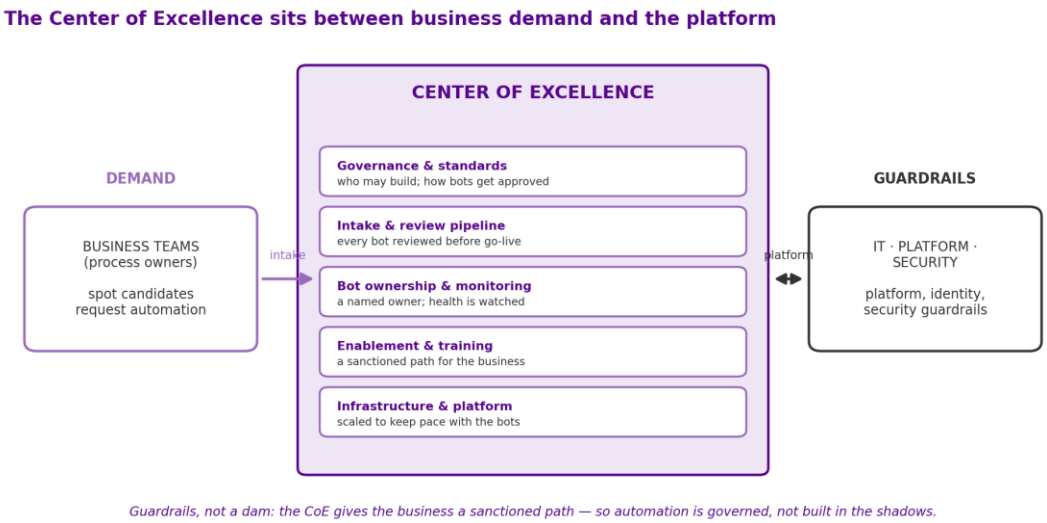


Figure 2. The Center of Excellence sits between the business (which supplies demand and process knowledge) and IT (which supplies the platform, identity, and security guardrails). It converts scattered requests into governed, owned, monitored automation.

The first function is governance and standards — the rules for who may build an automation, how it must be built, and how it is approved before it goes anywhere near production. This is the direct antidote to bot sprawl: a bot that has passed a review is not a shadow bot. Tella, Brinker and Pezzini describe a company that made this the very first thing its new automation team built — an automation review pipeline through which every automation the business created had to pass before it went live (Tella et al., 2023). The pipeline is not bureaucracy for its own sake; it is the mechanism that lets a company say yes to far more builders without saying yes to chaos. The second function is intake and the review pipeline itself — the sanctioned front door through which demand arrives, is triaged, and is either built centrally, handed back to a trained business builder under supervision, or declined. A visible front door is what makes the shadow door unnecessary.

The third function is bot ownership and monitoring, and it is the one students most often forget, even though it is where the ethics of this week live. Every bot in production has a named owner and a watched pulse: someone is accountable for it, and something is checking, continuously, that it is doing what it should. Consider the week's ethics question in its sharpest form: if a bot produces incorrect outputs silently for weeks, who is accountable, and how should the liability for automated errors be structured? A programme with no ownership function has no answer — the bot has no owner, its outputs are not watched, and the silent error runs until a customer complains or an auditor finds it. A CoE's ownership-and-monitoring function is exactly the answer: it assigns the accountable human, it defines what 'healthy' looks like for each bot, and it designs the audit trail so that a wrong output is caught in days, not discovered in months. Automation does not remove accountability; it relocates it, and the CoE is where it comes to rest. The fourth function is enablement and training — turning willing people in the business into capable, supervised builders — and the fifth is infrastructure and platform — owning the plant that runs the bots and, crucially, scaling it in pace with the programme so that O2's lead-machine collapse does not happen to you. Keep those five in view: governance, intake, ownership and monitoring,

enablement, infrastructure. A CoE that neglects any one of them stalls in the way that function protects against.

Now walk O2's operating model against that template, because it is the anchor for the whole idea and because its most instructive choice is one most companies get wrong. O2 ran its entire scaled programme — fifteen core processes, more than a hundred and sixty robots — with a team of just three in-house developers (Lacity and Willcocks, 2016). More striking is where those people came from and who owned the work: the programme lived in the Back Office, not in IT. Its builders were process-improvement staff reassigned to automation, which meant the head of the Back Office could move people from improving a process to automating one with, in effect, no change to the budget and no dependency on IT's development queue. That single organisational fact is what made O2's business case beat the alternative. When the company compared RPA against its existing business-process-management system on the same two processes, the technologies performed comparably; the financial gap came almost entirely from resourcing. The management system needed scarce IT developers and formal project teams and would have taken around three years to pay back; RPA, run by reassigned business people, paid back in roughly ten months (Lacity and Willcocks, 2016). The lesson is not that one tool is better than another. It is that a scaled automation programme is an operating-model decision before it is a technology decision — who owns it, who builds it, and against whose budget it runs.

Deloitte's survey puts the same idea in the language of the wider field and adds a distinction you will need. A classic Center of Excellence, in Deloitte's account, focuses on the high-value automations and the end-to-end process redesign that the whole company cares about, and it exists to drive the governance protocols under which more local automation can safely happen (Deloitte, 2022). That is the crucial word: the CoE governs; it does not personally build everything. It sits, as Figure 2 shows, between the business that supplies demand and knowledge and the IT function that supplies the platform, identity, and security guardrails, converting scattered requests into governed, owned, monitored automation. Deloitte also names the thing a CoE is often confused with — citizen-led development, where trained non-technical employees build their own low-complexity automations — and is careful to say it is not a replacement for a CoE but a complement that only works under a CoE's governance (Deloitte, 2022). Hold that distinction lightly for now; the machinery that makes safe citizen development possible is a later week's subject, and we preview it below rather than teach it here.

The CoE's five functions	What it does	Anchored exemplar
Governance & standards	Sets who may build, how, and how a bot is approved before production — the direct antidote to sprawl	Tella et al. (2023): every automation passes an 'automation review pipeline' before it goes live
Intake & review pipeline	A sanctioned front door for demand — triage, then build centrally, hand to a supervised builder, or decline	A visible front door makes the shadow door unnecessary (Tella et al., 2023)
Bot ownership & monitoring	Every bot has a named owner and a watched pulse; the audit trail catches a silent error in days	The answer to the week's ethics question — accountability is relocated, not removed
Enablement & training	Turns willing business people into capable, supervised builders	O2 ran the whole programme with 3 reassigned Back-Office builders (2016)

The CoE's five functions	What it does	Anchored exemplar
Infrastructure & platform	Owns the plant and scales it in pace with the programme	O2's lead-VM collapse is what this function exists to prevent (2016)

The CoE's job is to make automation a governed capability, not a pile of scripts. Its five functions — governance and standards, intake and review, bot ownership and monitoring, enablement, and infrastructure — each answer one way a programme stalls. O2 ran 160+ bots with three reassigned Back-Office builders and beat its alternative on operating model, not technology: who owns automation, who builds it, and against whose budget it runs. Accountability is relocated to the CoE, not removed by automation.

Scarcity Says Hoard the Experts; Scale Says Multiply Them

There is a mindset underneath the operating model, and if you miss it you will build a CoE that is just a bottleneck with a nicer name. Tella, Brinker and Pezzini frame it as a contrast between a scarcity mindset and a scale mindset, and the arithmetic they use to motivate it is worth carrying because a board will feel it immediately. Imagine a mid-size company with a couple of hundred applications and a thousand employees, each touching a handful of processes a day. The number of automatable processes runs into the thousands, most of them small and unnamed, living only in the heads of the people who do them. Now suppose a central team builds automations one at a time. Even at a brisk pace, the queue is a lifetime of work; the authors put the timeline to automate everything through a single specialist team at over eighty years (Tella et al., 2023). The point of the number is not its precision. It is that you cannot scale an automation programme by adding specialists, because there will never be enough of them, and treating scarce technical people as the only ones allowed to build is a scarcity mindset that guarantees the ninety-nine percent of processes never gets touched.

The scale mindset is the opposite move: instead of hoarding a scarce resource, you multiply who can use it, safely, under guardrails. Tella and his co-authors reach for an image that captures the fear and the answer at once. Picture a river held back by a dam. Remove the dam and you get a flood; leave it in place and the water never flows. The move is neither — it is to remove the dam and install guardrails that channel the current, so innovation flows fast but within banks (Tella et al., 2023). Applied to a company: you do not lock automation inside a tiny team (the dam), and you do not throw the tools open to everyone with no oversight (the flood). You build the guardrails — which is precisely what the CoE's governance, intake, and review functions are — and then you let far more people build within them. Read against Figure 2, this reframes what the CoE is for. It is not the company's automation factory; it is the company's automation steward. The authors put the shift crisply: central IT stops being an automation factory that the business queues up behind and becomes a service provider that enables the business to build. That is not a demotion of IT — it raises IT's standing, because stewarding the whole company's automation is a larger job than personally writing every bot.

This is also, directly, the answer to the second study question's harder half: how does a CoE prevent shadow IT? Not by policing it. Shadow IT is what people do when the sanctioned path is closed or too slow — it is a symptom of scarcity, of a central team that is a dam. The CoE drains

the shadow by opening a better-lit path than the shadow one: a review pipeline that says yes quickly, training that makes people competent, and support that makes the sanctioned way the easy way. Tella's account of the company that built its CoE this way is instructive precisely because the team resisted the urge to open a request queue and start taking build orders. It ran a class instead, licensed the graduates to build, and put every build through the review pipeline — and within months a large majority of the company's automations were being built by people who were not deep technical specialists, with the relationship between IT and the business repaired rather than strained (Tella et al., 2023). The CoE prevented shadow IT by making itself unnecessary to route around.

Two things here are previews, not this week's lesson, and you should hand them forward cleanly rather than reach for them now. The technology that makes safe, large-scale citizen development actually work — the platform and controls beneath democratisation — is a later subject; in Tella's own structure it is the architectural underpinning that follows the scale-mindset chapter, and in this course its fuller treatment comes with the agent-design and governance work later in the term. And the deeper questions of what to do when the tools become generative — when someone can describe an automation in words and have it built — belong to Modules 3 and 4. For Week 5, hold the mindset and the guardrails; the machinery underneath comes later.

A CoE that hoards is a bottleneck; a CoE that stewards is a multiplier. The scarcity mindset treats scarce specialists as the only ones allowed to build, so the ninety-nine percent of processes never gets automated. The scale mindset removes the dam and installs guardrails — the CoE's governance and review — so far more people build safely. That is also how a CoE prevents shadow IT: not by policing it, but by opening a better-lit path than the shadow one. The platform beneath large-scale citizen building is previewed for Week 8, not taught here.

Automate Alongside the Workforce, Not Instead of It

A scaled automation programme is a decision about people as much as about bots, and this section makes explicit a question Week 4 planted and promised would grow louder: even for a rules-based system, where should a human stay in the loop on purpose? The comfortable reading of 'rules-based' is 'no human needed', and it is a trap. Two bodies of thinking sharpen the answer — one about how automation and people fit together in an operation, one about how automation and people fit together across the whole design.

Westerman, Bonnet and McAfee, in the operations chapter of Leading Digital, give you the frame for the design question. They argue that managers have long treated a set of operational choices as painful either/or trade-offs, and that digital technology, used well, breaks the trade-off into a both/and. The one that matters most here is standardizing versus empowering. The old assumption was that standardising work — and then automating it — necessarily deskills and disempowers the people who did it. Westerman and his co-authors show companies for whom it did the opposite: automating the routine freed people into the judgement-heavy work only people can do. Their cleanest example is a paint company that moved routine order-taking from field salespeople into a standardised, centralised system — and, in doing so, turned those salespeople

from order-takers into relationship managers, doing more valuable work than the routine had ever allowed (Westerman, Bonnet and McAfee, 2014). The point for your programme is that ‘automate alongside the workforce, not instead of it’ is not a slogan you bolt on at the end; it is a design choice you make when you decide which part of a role the bot takes and what the person is freed to do instead.

The same authors add a second idea that reframes what a CoE is really building. They observe that operational advantage is peculiarly hard for competitors to copy, because it lives not in any single tool but in the way processes, skills, and information are tuned to work together as one machine — the workings are hidden even when the results are visible (Westerman, Bonnet and McAfee, 2014). Read that against the CoE. A pile of bots is easy to copy; anyone can license the same tool. A Center of Excellence — the people, the standards, the review pipeline, the accumulated judgement about which processes reward automation and which do not — is exactly the tuned, hidden capability that is hard to copy. When you argue for a CoE to your board, this is the argument that lands: you are not asking to buy bots, which any competitor can also buy; you are asking to build a capability they cannot simply purchase.

Tella and his co-authors give you the operational side of the human question, in a taxonomy that will also serve you as you preview the rest of the course. They distinguish three approaches to automating work. Task-based automation takes a single discrete step and automates it in isolation — the Week 4 move, useful and limited. Straight-through processing runs a whole flow end to end with no human touch at all, which is right for high-volume, time-critical work like payments but wrong for most processes, because most processes still need a person somewhere — to handle an exception, exercise judgement, or provide the human contact a customer actually wants. Their third approach, orchestration, coordinates people, software, and data across an end-to-end process, deliberately keeping people in the parts where their strengths matter most and letting the machinery carry the rest (Tella et al., 2023). One clarification you must carry, because the same word is about to mean three different things in this course: Tella’s ‘orchestration’ is an automation-tooling idea — how you weave tasks, systems, and people into one flow. It is not the same as the company roadmap phase you will meet in Week 11, and the deep machinery Tella builds it on — event-driven architecture and an enterprise automation platform — is Module 3 and 4 territory that this note previews and does not teach. For Week 5, take from the taxonomy only its human lesson: even at the pinnacle of automation, the mature design keeps people where judgement lives.

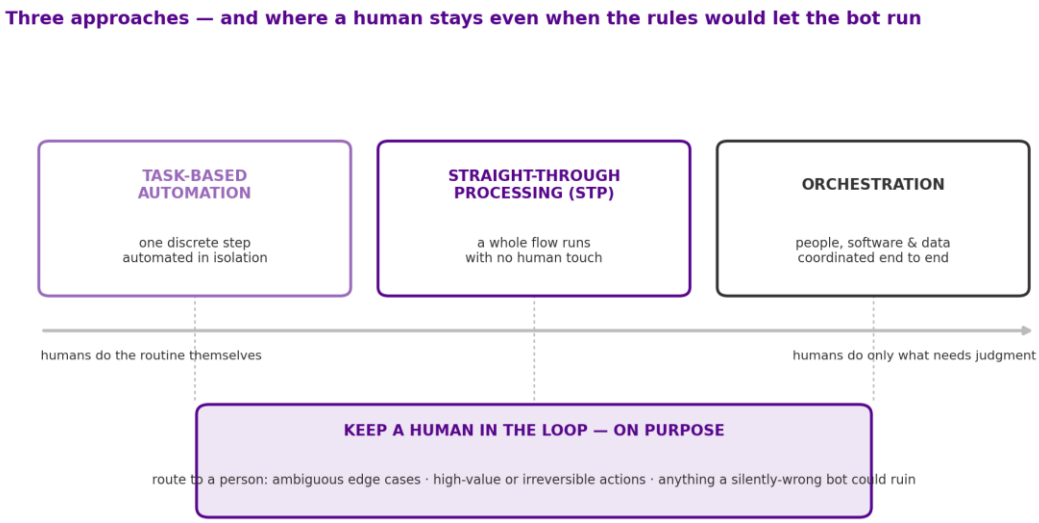


Figure 3. Task-based automation, straight-through processing, and orchestration differ in how much runs without a person — but in all three the mature design deliberately routes certain cases to a human.

Now answer the study question directly: when should a human be intentionally integrated into an automated workflow, even for a rules-based system? Route a case to a person whenever the cost of the bot being confidently wrong exceeds the cost of a human touch. Concretely, that means the ambiguous edge cases where the rule does not really fit — O2's multiple-iPhone order is the archetype, a case that looked rule-covered and was not; the high-value or irreversible actions, where a single wrong output does damage that cannot be undone by re-running the bot; and, above all, anything a silently-wrong bot could quietly ruin over weeks before anyone notices, which loops straight back to the ethics question and the CoE's monitoring function. 'Rules-based' tells you the routine can be automated; it does not tell you the whole step should run unattended. Choosing where the human stays is part of designing the automation, not an afterthought to it — and in a scaled programme it is a standing governance decision the CoE owns, not a one-off call a builder makes and forgets.

O2 lets you see the workforce side of this at scale, and honestly. Its programme did not eliminate a department; the company estimated the people it freed in the hundreds, and — tellingly — as service requests were handled faster and more accurately, the volume of customer 'chase up' calls fell by more than eighty percent, which in turn reduced the load on the front office as well as the back (Lacity and Willcocks, 2016). Some of those freed people were redeployed as the business grew. That is the genuine version of 'automate alongside the workforce' — and Week 4 taught you to be sceptical of the phrase, because 'redeployment to higher-value work' is sometimes a real commitment and sometimes a convenient narrative a firm tells to avoid a harder conversation. Hold the question open rather than closing it with a slogan; Week 13 returns to the human side of automation in full, and your Board Memo §7 will have to answer it for your own company rather than wave at it. For this week, plant the flag: a scaled programme is designed around people from the start, or it is designed to fail them.

‘Rules-based’ does not mean ‘no human needed’. Automating the routine can empower people into judgement work rather than deskill them (Westerman et al., 2014), and a CoE is the hard-to-copy capability, not the bots. Keep a human in the loop wherever a confidently-wrong bot costs more than a human touch: ambiguous edge cases, high-value or irreversible actions, and anything a silent error could ruin. The workforce obligation is real — hold it open; Week 13 answers it in full.

Two Telecoms, Two Kinds of Evidence

You have read O2 in pieces throughout this note; now assemble it as one story, because its arc is the arc of the week. Then set beside it a second telecom that looks superficially similar and is a completely different kind of evidence — and learning to tell those two kinds apart is itself a board skill.

The anchor — Telefónica O2 (Lacity and Willcocks, 2016)

The problem. O2 is a large UK mobile operator whose back office, like most, faced relentless pressure to do more work for less money. Through the 2000s it pursued the standard cost levers — offshoring, then process elimination and simplification — and reached the point where offshoring had little left to give and the mandate became blunt: cut headcount, response time, and failure-driven customer calls each by half. Automation surfaced as the next lever.

What happened. O2 ran a deliberately small proof of concept — two high-volume, low-complexity processes, automated in about two weeks — to answer three questions its board would ask of any new tool: will it integrate with our systems without breaking them, will it deliver quality, and will it pay back (Lacity and Willcocks, 2016)? It did, so emphatically that the bots’ volume tripped the company’s own fraud alarms and the programme’s sponsor was briefly suspected of an intrusion. Then O2 did the disciplined thing and tested RPA against its incumbent business-process-management system on comparable processes; the two automated the work similarly well, but RPA won decisively on the business case — roughly ten months to pay back against about three years — and, as we saw, the reason was the operating model, not the technology: RPA was run by reassigned Back-Office people with no draw on IT’s scarce developers. From there the programme scaled in waves — twenty robots, then seventy-five, then more than a hundred and sixty — run throughout by a team of three in-house developers, automating fifteen core processes and reaching about thirty-five percent of all back-office transactions, at four to five hundred thousand transactions a month, a twelve-month payback, and a three-year return on investment the case puts at between six hundred and fifty and eight hundred percent (Lacity and Willcocks, 2016). The scaling was not frictionless — recall the lead-machine collapse and the sixteen weeks of infrastructure re-engineering — which is exactly why the case is a scaling lesson and not a fairy tale.

The lesson. The value landed on the three threads this course always watches. On Operations and Supply Chain, the repetitive rules-based work was carried by bots at a fraction of the cost. On Cash, turnaround times fell from days to minutes, compressing cash conversion. On Customer, the eighty-percent drop in chase-up calls meant the service was fast and accurate enough that customers no longer needed to phone. And the five lessons O2 itself drew read like

a charter for a Center of Excellence: sequence process elimination and improvement before automation; give robots more explicit instructions than humans because they lack common sense; grow the infrastructure in pace with the programme; choose the sourcing model deliberately; and bring IT in early rather than pioneering behind its back (Lacity and Willcocks, 2016). That last one is the governance moral of the fraud-alarm story: the pilot succeeded, but it nearly cost its sponsor his job precisely because there was no sanctioned path — which is the case, in miniature, for building one.

The illustration — Orange Spain's 'Robot Factory' (UiPath vendor case study)

Now set beside O2 a second telecom, and change how you read. The Orange Spain case describes a Center of Excellence — a 'Robot Factory' — that the vendor reports grew from thirteen people and zero bots into a team of around a hundred deploying more than four hundred bots across the company, saving, it says, over thirty-four million euros in a little over two years, while training employees across the company to build automations. Read as a growth arc, this illustrates two things this week teaches: the same pilot-to-enterprise trajectory O2 lived, and the human dimension — hundreds of employees trained as builders is the scale mindset made visible, automation built alongside the workforce rather than dropped on it.

But read it as evidence and everything changes, and this is the deliberate lesson of pairing the two cases. Orange Spain is a UiPath customer story: it is marketing, produced by the vendor whose product it praises, to sell that product. Every figure in it — the four hundred bots, the thirty-four million euros, the hundred-strong team — is a claim the vendor is making, not a number independently verified by researchers, and you should treat it that way. That does not make the figures false; it makes them uncorroborated, which is a different thing your board must understand. The discipline is threefold: never let a vendor number carry a load-bearing claim in your argument; corroborate any figure you actually need from an independent source before you rely on it; and capture the source's title, publisher, and date so that even if the link dies, the provenance stays traceable. O2 and Orange Spain teach the same scaling lesson, but they are not the same kind of evidence — one is peer-reviewed research, one is a sales asset — and a board-ready analyst is the person who can hold both, use each for what it is good for, and never confuse the two. That is why the two telecoms travel together in this note.

Use the anchor to establish, the vendor case to illustrate — and never confuse the two.
Telefónica O2 is peer-reviewed research: its figures — pilot to 160+ bots, three builders, ~35% of transactions, days-to-minutes, 650–800% ROI — can carry your argument. Orange Spain's 'Robot Factory' is vendor marketing: its figures are claims, useful only to illustrate the growth arc and the certified-builder workforce, never to carry a load-bearing point. Corroborate what you rely on; capture provenance so a dead link stays traceable.

Where This Points: The Threads, the Labs, and the Capstone

Pull the week together on the three axes this course always uses. On the threads, scaling and governance sit first on Operations and Supply Chain — the programme finds most of its bots in operational, rules-based work — but you have watched the payoff land on Cash (O2's days-to-minutes turnaround) and on Customer (the collapse in chase-up calls). When you design your

company's operating model, tag the value by thread, because a board funds a thread outcome, not an org chart.

On the labs, this week is the concept that Lab 2 makes physical. In Week 6 you will take the stable-rules decision point you found in Lab 1 — your RPA candidate — and build it in Make, and then you will deliberately break it. That break is not a mishap; it is this week's selector-instability lesson delivered at one-bot scale, so the brittleness you read about here stops being abstract. And the one-page post-mortem Lab 2 asks for — what broke, what human oversight you would build in — is the CoE's bot-ownership-and-monitoring function practised on a single bot before you argue for it across a company. Keep the lab spine straight, unchanged since Week 4: the stable-rules decision point is your RPA candidate and your Lab 2 target; the judgment decision point is not for RPA — it branches off to become your AI-agent target — you pin it down in Week 7 and build it in Lab 3 (Week 9).

On the capstone, your deliverable is the operating-model half of Board Memo §4 (Automation Journey): a draft Center-of-Excellence and operating-model section for your company. Week 4 gave §4 its first half — the RPA-candidate scan, which fills the Automation-opportunity slide. This week gives §4 its second half — how automation would be owned, governed, and run at scale — which fills the Automation-journey slide, S7. Both are still Diagnose-phase work: you are not yet proposing the intelligent, agentic future; you are showing the board that you know how a company would actually run automation, not just where a bot could go. Lab 2's working automation and post-mortem complete §4's evidence.

Two directions are deliberately left for later, and you should hand them forward rather than reach for them. The move from rules to genuine machine judgement — intelligent automation — is Week 7; the agent's SPAR architecture is Week 8; and you build the agent in Lab 3, Week 9. The platform and controls beneath safe large-scale citizen building come later in the term, not this week. The deeper redesign toward event-driven operations, and the strategic roadmap a board commits to, are Modules 3 and 4 — and note again that this course uses 'orchestrate' in more than one sense: Tella's tooling sense is not the roadmap phase you will build in Week 11, and this note deliberately avoids the roadmap's language so the two never blur. This week teaches only the second rung fully: how a programme scales, who governs it, and where the human stays.

This week designs how automation is run; the next steps make it intelligent and, later, strategic. Tag your operating model's value by thread — boards fund outcomes, not org charts. Lab 2 builds your Lab 1 stable-rules candidate in Make and breaks it, making selector instability and the ownership discipline concrete; the judgment point is handed to AI (Week 7, Lab 3). Your deliverable is the CoE / operating-model half of Board Memo §4 — the Automation-journey slide, S7 — still Diagnose-phase work.

Writing Your CoE and Operating-Model Section

Turn all of this into the deliverable, because an operating model is only useful if it is written in a form a board could actually staff. Do not draw an aspirational org chart. Design the smallest governing arrangement that would let your company run automation past the pilot — sized

honestly to your archetype — and write it as a short section a board could act on. Six questions structure it, and they map one-to-one onto the CoE's five functions plus the human question.

First, who owns automation? Name the home — and remember O2's lesson that the most consequential choice is whether it lives with the business or with IT, and against whose budget it runs. Second, what is the sanctioned path — how does a new automation request arrive, get reviewed, and get approved before it goes live? This is your answer to bot sprawl and shadow IT. Third, who owns each bot in production, and how is its health watched — so that a silent error is caught in days, not discovered by a customer? This is where you answer the week's accountability question explicitly. Fourth, how do you enable more builders safely — do you train business people, and if so under what guardrails? Fifth, how does the infrastructure scale so your programme does not hit O2's lead-machine wall? And sixth, running through all of them: where do you deliberately keep a human in the loop, and what does the company owe the people whose routine work the programme automates?

Scale the answer to your archetype rather than copying O2's. For the fifty-person services firm on HubSpot and Notion, the 'Center of Excellence' may be one named owner, a one-page rule for what may be automated and who signs off, and a shared list of every automation with an owner beside it — that is a governing model, small but real. For the mid-market manufacturer on NetSuite, it may be a two-person function that reviews and owns bots and trains a handful of business builders. For the CPG enterprise on SAP, it may look much like O2's. Keep it honest and short: a board is more convinced by a small operating model it believes you could staff on Monday than by a grand one it knows you cannot. That written section, with Lab 2's working bot and post-mortem beside it, is your Board Memo §4 — and it is worth more than a longer plan precisely because it shows you know how automation is run, not just where it could go.

Design the smallest governing model your company could actually staff. Answer six questions: who owns automation (and against whose budget), what the sanctioned build path is, who owns and monitors each bot, how you enable more builders safely, how infrastructure scales, and where the human stays. Size it to your archetype — one named owner and a written rule is a real model for a fifty-person firm. A small model a board believes beats a grand one it doesn't. This is Board Memo §4.

Study Questions

Bring reasoned answers; they are designed to be argued with the cases and frameworks, not recited.

1. What is 'selector instability', and how does an unexpected system update break an RPA script? Explain the mechanism in your own words, then use the O2 iPhone example — or a step from your own Lab 1 map — to show why 'the bot kept running and did the wrong thing' can be worse than 'the bot stopped'.

2. What is the primary managerial function of an Automation Center of Excellence, and how does it prevent shadow IT? Answer both halves: what a CoE governs (not just builds), and why opening a sanctioned path drains the shadow better than policing it does. Ground your answer in O2's Back-Office-owned model and Tella's scale mindset.

3. When should a human be intentionally integrated into an automated workflow, even for a rules-based system? Give the general rule — route to a person when a confidently-wrong bot costs more than a human touch — and three concrete triggers. Connect your answer to the week's ethics question about a bot that is silently wrong for weeks, and note what you expect Week 13 to add.

Running Glossary

New this week, plus the carried terms this note builds on.

Center of Excellence (CoE). A small, permanent function that owns how automation is done across a company — setting standards, running the intake and review pipeline, owning and monitoring bots, enabling builders, and owning the platform. It governs automation as a capability rather than building every bot itself.

Operating model (for automation). The organisational design of a programme: who owns automation, who builds it, against whose budget it runs, and how it is governed. O2's case shows this is a bigger determinant of success than the choice of tool.

Bot ownership. The discipline of giving every production bot a named accountable owner and a monitored health signal, with an audit trail that catches a wrong output in days. Where accountability comes to rest once a task is automated.

Selector instability. A bot addresses screen elements by selectors recorded when it was built; a routine system update that moves, renames, or re-wraps an element breaks those selectors, so the script fails or silently does the wrong thing. The mechanical form of RPA's brittleness.

Integration drift. The gradual failure that accumulates as the many systems a programme's bots ride on evolve independently over time, so stable seams become moving ones and maintenance grows faster than the bot count.

Bot sprawl (shadow automation). The uncontrolled proliferation of unowned, unmonitored automations built off to the side when there is no sanctioned path — fragmentation with bots on top. The organisational failure a CoE exists to prevent.

Scale mindset (vs scarcity mindset). Treating the ability to build automation as something to multiply under guardrails rather than a scarce resource to hoard inside a specialist team. 'Remove the dam, install guardrails' (Tella et al., 2023).

Shadow IT / rogue IT. Technology or automation adopted by the business without IT's sanction; in its worse form (rogue IT), business data moved or transformed in unsanctioned, unwatched ways. A symptom of a closed or slow sanctioned path.

Guardrails. The standards, intake review, and controls a CoE puts around building so that many people can automate safely — the mechanism that lets a company embrace scale without embracing chaos.

Straight-through processing (STP). Running a whole flow end to end with no human touch; right for high-volume, time-critical work, wrong for most processes, which still need a person for exceptions and judgement (Tella et al., 2023).

Orchestration (automation-tooling sense). Coordinating people, software, and data across an end-to-end process, keeping people where their strengths matter (Tella et al., 2023). Distinct

from the Week-11 roadmap phase of the same name and from Westerman's operational 'orchestrating' — do not conflate the three.

Human-in-the-loop. A deliberate design choice to route certain cases to a person even in a rules-based system — ambiguous edge cases, high-value or irreversible actions, and anything a silently-wrong bot could ruin. A standing governance decision the CoE owns.

RPA candidate criteria. The three tests a step must pass to be worth a bot: repetitive, rules-based, and stable. (Introduced Week 4; the stable-rules decision point that passes them is the Lab 2 target.)

Leverage gap. The distance between a backbone that works technically and a backbone that delivers business value. Closing it, layer by layer, is the work of the course. (Introduced Week 1.)

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